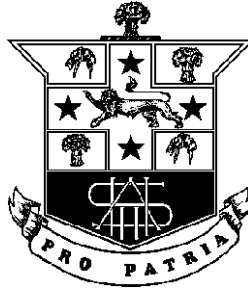


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION 2

2006

YEAR 12

HALF YEARLY EXAMINATION

Examiners ~ Z Pethers, G Rawson

GENERAL INSTRUCTIONS

- | | |
|---|--|
| <ul style="list-style-type: none">• Reading Time – 5 minutes.• Working Time – 2 HOURS.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains five (5) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started in a new booklet.• This examination paper must NOT be removed from the examination room. |
|---|--|

STUDENT NAME: _____

TEACHER: _____

Question 1 (15 marks) Start a NEW answer booklet

Marks

- (a) Find all pairs of integers x and y that satisfy $(x + iy)^2 = 33 + 56i$ **2**
- (b) (i) Let $z = \cos \frac{\pi}{8} + i \sin \frac{\pi}{8}$. Find z^8 . **2**
- (ii) Plot on the Argand diagram, all complex numbers that are solutions of $z^8 = -1$. **2**
- (c) Find all the roots of the equation $5z^3 - 31z^2 + 56z - 10 = 0$, given that $(3 + i)$ is one of the roots. **3**
- (d) (i) If $z_1 = 2 + 4i$, sketch the locus of $(\alpha) |z - z_1| = 3$ **1**
- $(\beta) \arg(z - z_1) = \frac{\pi}{4}$ **1**
- (ii) Find the area specified by $|z - z_1| \leq 3$ and $\arg(z - z_1) \leq \frac{\pi}{4}$ **2**
- (iii) If $|z - z_1| = 3$, find the greatest value of $|z - 7|$. **2**

Question 2 (15 marks) Start a NEW answer booklet

		Marks
(a)	(i) Suppose the polynomial $P(x)$ has a double root at $x = \alpha$. Prove that $P'(x)$ also has a root at $x = \alpha$.	2
	(ii) The polynomial $P(x) = x^4 + ax^3 + bx + 21$ has a double root at $x = 1$. Find the values of a and b .	3
	(iii) Factorise the polynomial $P(x) = x^4 + ax^3 + bx + 21$ over the real numbers.	2
(b)	If $\frac{2x+31}{(x-1)^3(x+2)} = \frac{a}{(x-1)} + \frac{b}{(x-1)^2} + \frac{c}{(x-1)^3} + \frac{d}{(x+2)}$, find the values of a, b, c, d .	3
(c)	The equation $x^4 + 3x^3 + 5x^2 - 7x + 2 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find the equation with roots $\alpha^2, \beta^2, \gamma^2, \delta^2$.	3
(d)	The equation $x^3 - 4x^2 + 5x + 2 = 0$ has roots α, β, γ . Find the value of $\alpha^2 + \beta^2 + \gamma^2$.	2

Question 3 (15 marks) Start a NEW answer booklet

Marks

Consider the hyperbola: $x^2 - 3y^2 = 6$.

- | | | |
|-----|--|----------|
| (a) | Find the eccentricity of the hyperbola. | 1 |
| (b) | Find the coordinates of the foci and the equations of the directrices and asymptotes of the hyperbola. | 3 |
| (c) | Sketch the graph of the hyperbola showing clearly all of the above features and the intercepts on the coordinate axes. | 2 |
| (d) | Use differentiation to derive the equations of the tangent and normal to the hyperbola at the point $P(3, -1)$. | 3 |
| (e) | The tangent meets the transverse axis in M and normal meets the conjugate axis in N . Find the distance MN . | 1 |
| (f) | Determine the angle between the asymptotes. | 2 |
| (g) | Find the equations of the tangents to $x^2 - 3y^2 = 6$ which are parallel to the line $2x - y = 7$. | 3 |

Question 4 (15 marks) Start a NEW answer booklet

Marks

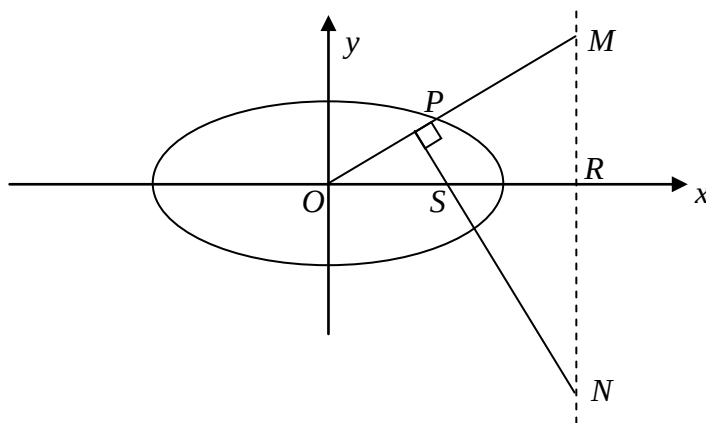
(a) (i) Find the equation of the ellipse with parametric equations $x = 2\cos\theta$ and $y = \sin\theta$. 2

(ii) Sketch the ellipse, showing its foci and directrices. Show the auxillary circle on your sketch and state its equation. 2

(b) Using differentiation, find the equation of the normal to the curve whose parametric equations are $x = 2\sec\theta$ and $y = 3\tan\theta$ at the point P where $\theta = \frac{\pi}{3}$. 3

(c) Prove that the condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 + b^2$. 3

(d) $P(a\cos\theta, b\sin\theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre O .



OP produced meets a directrix in M , and the perpendicular from the corresponding focus onto OM meets the same directrix in N . If R is the foot of the directrix, prove that the product of RM and RN is independent of the position of P .

5

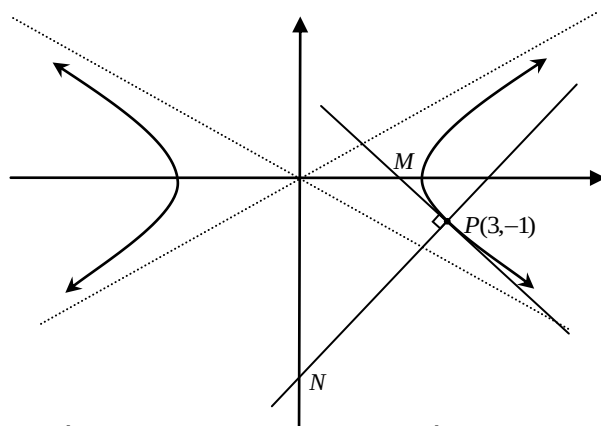
Question 5 (15 marks) Start a NEW answer booklet

Marks

- (a) Consider the polynomial equation $x^4 + ax^3 + bx^2 + cx + d = 0$, where a, b, c , and d are all integers. Suppose the equation has a root of the form ki , where k is real, and $k \neq 0$.
- (i) State why the conjugate $-ki$ is also a root. 1
- (ii) Show that $c = k^2a$. 2
- (iii) Show that $c^2 + a^2d = abc$. 2
- (iv) If 2 is also a root of the equation, and $b = 0$, show that c is even. 2
- (b) Let $\rho = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$.
- The complex number $\alpha = \rho + \rho^2 + \rho^4$ is a root of the quadratic equation $x^2 + ax + b = 0$, where a and b are real.
- (i) Prove that $1 + \rho + \rho^2 + \dots + \rho^6 = 0$. 2
- (ii) The second root of the quadratic equation is β . Express β in terms of positive powers of ρ . Justify your answer. 2
- (iii) Find the values of the coefficients of a and b . 2
- (iv) Deduce that $-\sin \frac{\pi}{7} + \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{2}$. 2

Year 12	Mathematics Extension 2	HY Examination 2006												
Question No. 3	Solutions and Marking Guidelines													
Outcomes Addressed in this Question														
E3	uses the relationship between algebraic and geometric representations of conic sections													
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections													
Outcome	Solutions	Marking Guidelines												
(a) E3, E4	$x^2 - 3y^2 = 6$ $\frac{x^2}{6} - \frac{y^2}{2} = 1 \quad \Rightarrow \quad b^2 = a^2(e^2 - 1)$ $e = \sqrt{\frac{b^2}{a^2} + 1}$ $= \sqrt{\frac{2}{6} + 1}$ $= \frac{2}{\sqrt{3}}$	<u>1 mark</u> : correct answer												
(b) E3, E4	<table><thead><tr><th>Foci</th><th>Directrices</th><th>Asymptotes</th></tr></thead><tbody><tr><td>$(\pm ae, 0)$</td><td>$x = \pm \frac{a}{e}$</td><td>$y = \pm \frac{b}{a}x$</td></tr><tr><td>$= \left(\pm \sqrt{6} \times \frac{2}{\sqrt{3}}, 0 \right)$</td><td>$x = \pm \frac{\sqrt{6} \times \sqrt{3}}{2}$</td><td>$y = \pm \frac{\sqrt{2}}{\sqrt{6}}x$</td></tr><tr><td>$= (\pm 2\sqrt{2}, 0)$</td><td>$x = \pm \frac{3\sqrt{2}}{2}$</td><td>$y = \pm \frac{1}{\sqrt{3}}x$</td></tr></tbody></table>	Foci	Directrices	Asymptotes	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{a}x$	$= \left(\pm \sqrt{6} \times \frac{2}{\sqrt{3}}, 0 \right)$	$x = \pm \frac{\sqrt{6} \times \sqrt{3}}{2}$	$y = \pm \frac{\sqrt{2}}{\sqrt{6}}x$	$= (\pm 2\sqrt{2}, 0)$	$x = \pm \frac{3\sqrt{2}}{2}$	$y = \pm \frac{1}{\sqrt{3}}x$	<u>3 marks</u> : all correct OR <u>2 marks</u> : 2 correct OR <u>1 mark</u> : 1 correct
Foci	Directrices	Asymptotes												
$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{a}x$												
$= \left(\pm \sqrt{6} \times \frac{2}{\sqrt{3}}, 0 \right)$	$x = \pm \frac{\sqrt{6} \times \sqrt{3}}{2}$	$y = \pm \frac{\sqrt{2}}{\sqrt{6}}x$												
$= (\pm 2\sqrt{2}, 0)$	$x = \pm \frac{3\sqrt{2}}{2}$	$y = \pm \frac{1}{\sqrt{3}}x$												
(c) E3, E4		<u>2 marks</u> : correct diagram with x-intercepts OR <u>1 mark</u> : correct diagram, but partially labelled												
(d) E3, E4	<table><tbody><tr><td>$x^2 - 3y^2 = 6$$2x - 6y \cdot \frac{dy}{dx} = 0$$\frac{dy}{dx} = \frac{x}{3y}$$= \frac{3}{-3}$$= -1 \text{ at } P(3, -1)$</td><td>Eq'n of tangent $y + 1 = -1(x - 3)$$x + y = 2$ Eq'n of normal $y + 1 = 1(x - 3)$$x - y = 4$</td></tr></tbody></table>	$x^2 - 3y^2 = 6$ $2x - 6y \cdot \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{x}{3y}$ $= \frac{3}{-3}$ $= -1 \text{ at } P(3, -1)$	Eq'n of tangent $y + 1 = -1(x - 3)$ $x + y = 2$ Eq'n of normal $y + 1 = 1(x - 3)$ $x - y = 4$	<u>3 marks</u> : correct solution OR <u>2 marks</u> : substantially correct OR <u>1 mark</u> : partially correct										
$x^2 - 3y^2 = 6$ $2x - 6y \cdot \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{x}{3y}$ $= \frac{3}{-3}$ $= -1 \text{ at } P(3, -1)$	Eq'n of tangent $y + 1 = -1(x - 3)$ $x + y = 2$ Eq'n of normal $y + 1 = 1(x - 3)$ $x - y = 4$													

Q.3 pg2
(e)
E3, E4



At M,
 $y = 0$ and $x + y = 2$
ie $x = 2$
so M is (2, 0)

At N,
 $x = 0$ and $x - y = 4$
ie $y = -4$
so N is (0, -4)

$$\text{Thus } MN = \sqrt{2^2 + 4^2} \\ = 2\sqrt{5} \text{ units}$$

1 mark : correct solution

(f)
E3, E4

Asymptote $y = \frac{1}{\sqrt{3}}x$ makes an angle

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} \text{ with the } x\text{-axis}$$

By symmetry, the (acute) angle between the asymptotes is $2 \times \frac{\pi}{6} = \frac{\pi}{3}$

$$\text{Alternatively, use } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \text{ with } \begin{cases} m_1 = \frac{1}{\sqrt{3}} \\ m_2 = -\frac{1}{\sqrt{3}} \end{cases}$$

2 marks : correct solution

OR

1 mark : substantially correct

(g)
E3, E4

Required equations have $m = 2$, ie they are of form $y = 2x + k$

Solving simultaneously with $x^2 - 3y^2 = 6$

$$x^2 - 3(2x + k)^2 = 6$$

$$\text{Gives } x^2 - 3(4x^2 + 4kx + k^2) - 6 = 0$$

$$11x^2 + 12kx + 3(k^2 + 2) = 0$$

Which must have equal roots for tangent

$$\text{i.e. } (12k)^2 - 4 \cdot 11 \cdot 3(k^2 + 2) = 0$$

$$12k^2 - 11(k^2 + 2) = 0$$

$$k^2 = 22$$

$$k = \pm\sqrt{22}$$

So equations of required tangents are $y = 2x \pm \sqrt{22}$

3 marks : correct solution

OR

2 marks : substantially correct

OR

1 mark : partially correct

Q.3 pg3

Alternate solution:

$$x^2 - 3y^2 = 6 \dots (1)$$

$$2x - 6y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{3y}$$

tangent must have
gradient = 2

$$\text{so } \frac{dy}{dx} = \frac{x}{3y} = 2$$

$$\text{ie } x = 6y$$

subbing this into (1)

$$x^2 - 3y^2 = 6$$

$$36y^2 - 3y^2 = 6$$

$$y = \pm \frac{\sqrt{2}}{\sqrt{11}}$$

$$\text{and this gives } x = \pm \frac{6\sqrt{2}}{\sqrt{11}}$$

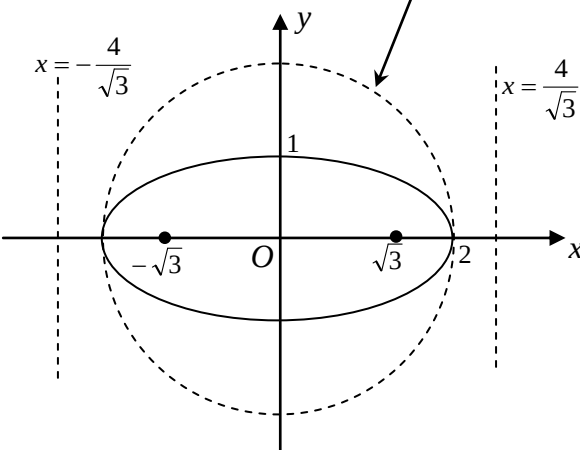
now, using

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\sqrt{2}}{\sqrt{11}} = 2 \left(x - \frac{6\sqrt{2}}{\sqrt{11}} \right)$$

$$y = 2x - \sqrt{22}$$

$$\text{also, } y = 2x + \sqrt{22}$$

Year 12	Mathematics Extension 2	HY Examination 2006
Question No. 4	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3	uses the relationship between algebraic and geometric representations of conic sections	
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections	
Outcome	Solutions	Marking Guidelines
(a)(i) E3, E4	$\left. \begin{matrix} x = 2 \cos \theta \\ y = \sin \theta \end{matrix} \right\} \Rightarrow a = 2, b = 1$ so ellipse has equation $\frac{x^2}{4} + y^2 = 1$	<u>2 marks</u> : correct solution OR <u>1 mark</u> : substantially correct
(a)(ii) E3, E4	$b^2 = a^2(1 - e^2) \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$ foci: $(\pm ae, 0) \Rightarrow \left(\pm 2 \times \frac{\sqrt{3}}{2}, 0 \right)$ ie $(\pm \sqrt{3}, 0)$ directrices: $x = \pm \frac{a}{e} = \pm \frac{2}{\frac{\sqrt{3}}{2}} = \pm \frac{4}{\sqrt{3}}$ equation of auxiliary circle is $x^2 + y^2 = 4$ 	<u>2 marks</u> : correct diagram with auxiliary circle OR <u>1 mark</u> : correct ellipse, partially labelled
(b) E3, E4	$\begin{aligned} x &= 2 \sec \theta & y &= 3 \tan \theta \\ \frac{dx}{d\theta} &= 2 \sec \theta \tan \theta & \frac{dy}{d\theta} &= 3 \sec^2 \theta \\ \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3 \sec^2 \theta}{2 \sec \theta \tan \theta} = \frac{3}{2 \sin \theta} \end{aligned}$	

Q.4 pg2

Now, when $\theta = \frac{\pi}{3}$

Normal has

$$m = -\frac{2\sin\theta}{3}$$

$$= -\frac{2 \cdot \frac{\sqrt{3}}{2}}{3}$$

$$= -\frac{\sqrt{3}}{3}$$

$$x = 2\sec\theta$$

$$= 2\sec\frac{\pi}{3}$$

$$= 2 \times 2$$

$$= 4$$

$$y = 3\tan\theta$$

$$= 3\tan\frac{\pi}{3}$$

$$= 3\sqrt{3}$$

$$y - y_1 = m(x - x_1)$$

$$y - 3\sqrt{3} = -\frac{\sqrt{3}}{3}(x - 4)$$

$$\sqrt{3}x + 3y - 13\sqrt{3} = 0$$

**(c)
E3, E4**

Tangent at $P(x_1, y_1)$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \dots \textcircled{1}$

To find condition for $y = mx + c$ to touch ellipse, we write this as $mx - y = -c$

$$-\frac{mx}{c} + \frac{y}{c} = 1 \dots \textcircled{2}$$

Both $\textcircled{1}$ and $\textcircled{2}$ represent the same tangent, So, comparing coefficients:

$$\frac{x_1}{a^2} = -\frac{m}{c}$$

and

$$\frac{y_1}{b^2} = \frac{1}{c}$$

$$x_1 = \frac{-a^2 m}{c}$$

$$y_1 = \frac{b^2}{c}$$

Subbing these values into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, we have:

$$\frac{a^4 m^2}{a^2 c^2} + \frac{b^4}{b^2 c^2} = 1$$

$$\frac{a^2 m^2}{c^2} + \frac{b^2}{c^2} = 1$$

$$c^2 = a^2 m^2 + b^2$$

Alternate solution:

Solving $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ and $y = mx + c$ gives:

$$\frac{x^2}{a^2} + \frac{(mx + c)^2}{b^2} = 1$$

$$b^2 x^2 + a^2 m^2 x^2 + 2a^2 mcx + a^2 c^2 - a^2 b^2 = 0$$

which must have one solution for tangent, ie $\Delta = 0$.

$$\Delta = 4a^4 - 4(a^2 m^2 + b^2)(a^2 c^2 - a^2 b^2) = 0$$

$\Downarrow$

$$c^2 = a^2 m^2 + b^2$$

3 marks : correct solution

OR

2 marks : substantially correct

OR

1 mark : partially correct

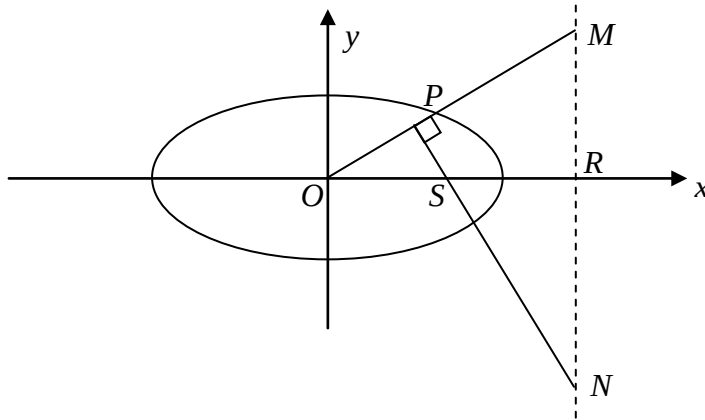
3 marks : correct solution

OR

2 marks : substantially correct

OR

1 mark : partially correct

(d)
E3, E4

Equation of OP is $y = \frac{b \sin \theta}{a \cos \theta} x$ and OP meets the directrix $x = \frac{a}{e}$ in M , thus, the coordinates of M are

$$x = \frac{a}{e}, \quad y = \frac{b \sin \theta}{a \cos \theta} \cdot \frac{a}{e} = \frac{b \sin \theta}{e \cos \theta}$$

$$\text{Hence, } RM = \left| \frac{b \sin \theta}{e \cos \theta} \right|$$

$SN \perp OP$ so SN has gradient $-\frac{a \cos \theta}{b \sin \theta}$ & passes through the focus $(ae, 0)$

$$\text{so has equation } y - 0 = -\frac{a \cos \theta}{b \sin \theta} (x - ae)$$

SN meets the directrix $x = \frac{a}{e}$ in N , thus, the

coordinates of N are

$$x = \frac{a}{e}, \quad y = \frac{-a \cos \theta}{b \sin \theta} \left(\frac{a}{e} - ae \right)$$

$$= \frac{-a^2(1-e^2)\cos\theta}{eb\sin\theta}$$

$$= \frac{-b^2 \cos \theta}{eb \sin \theta}, \quad \text{since } b^2 = a^2(1-e^2).$$

$$\text{Hence, } RN = \left| \frac{b \cos \theta}{e \sin \theta} \right|$$

$$\therefore RM \times RN = \left| \frac{b \sin \theta}{e \cos \theta} \right| \left| \frac{b \cos \theta}{e \sin \theta} \right| = \frac{b^2}{e^2}$$

which is independent of θ .

5 marks : correct solution

OR

4 marks : substantially correct

OR

3 marks : correctly finding the length of either RM or RN and partially correct solution thereafter.

OR

2 marks : correctly finding the length of either RM or RN

OR

1 mark : some evidence of an appropriate method of solution

Year 12	Mathematics Extension 2	HY Examination 2006
Question No. 5	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E2	chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings	
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving polynomials	
E9	communicates abstract ideas and relationships using appropriate notation and logical argument	
Outcome	Solutions	Marking Guidelines
(a)(i) E4	For polynomials with real coefficients, complex roots occur in conjugate pairs. So, if ki is a root, $-ki$ must also be a root.	<u>1 mark</u> : correct statement – must include reference to real coefficients
(a)(ii) E4	$P(x) = x^4 + ax^3 + bx^2 + cx + d$ $P(ki) = k^4 - ak^3i - bk^2 + cki + d = 0$ Equating imaginary parts: $-ak^3 + ck = 0$ $c = ak^2$ OR... $(x - ki)(x + ki) = x^2 + k^2 \text{ is a factor}$ so, $P(x) = x^4 + ax^3 + bx^2 + cx + d = (x^2 + k^2)(x^2 + Ax + B)$ Equating coefficients of x^3 : $a = A$ Equating coefficients of x : $c = Ak^2$ $\therefore c = ak^2$ OR... Let the roots of $P(x)$ be $ki, -ki, \alpha, \beta$ $\sum \alpha = \alpha + \beta = -a$ $\sum \alpha\beta\gamma = ki\alpha\beta - ki\alpha\beta + k^2\alpha + k^2\beta$ $= k^2(\alpha + \beta) = -c$ $\therefore k^2(-a) = -c$ $c = ak^2$	<u>2 marks</u> : correct solution OR <u>1 mark</u> : substantially correct
(a)(iii) E4	From $P(ki) = 0$ in (ii), equate real parts ie $k^4 - bk^2 + d = 0$ & noting $k^2 = \frac{c}{a}$ $\frac{c^2}{a^2} - b\frac{c}{a} + d = 0$ $c^2 - abc + ad = 0$ $c^2 + ad = abc$	<u>2 marks</u> : correct solution OR <u>1 mark</u> : substantially correct

Q.5 pg2
(a)(iv)
E2, E4

$$b = 0 \Rightarrow P(x) = x^4 + ax^3 + cx + d = 0$$

roots are $ki, -ki, 2, \alpha$

$$\sum \alpha = 2 + \alpha = -a$$

$$\sum \alpha\beta = k^2 + 2\alpha = 0$$

$$\alpha = -(a + 2)$$

$$k^2 = 2\alpha$$

$$\text{Noting } k^2 = \frac{c}{a} \Rightarrow \frac{c}{a} = -2\alpha$$

$$c = -2\alpha$$

$$c = 2a(a + 2)$$

a is an integer, $\therefore c$ is even

OR...

$$P(2) = 0 \Rightarrow 16 + 8a + 4b + 2c + d = 0$$

$$\text{since } b = 0: 16 + 8a + 2c + d = 0$$

$$\text{Now, } c^2 + a^2d = abc$$

$$b = 0 \Rightarrow a^2d = -c^2$$

$$d = \frac{-c^2}{a^2}$$

$$\therefore 16 + 8a + 2c - \frac{c^2}{a^2} = 0$$

$$c^2 - 2a^2c - 16a^2 - 8a^3 = 0$$

Quadratic Formula gives

$$c = 2a(a + 2) \quad \text{or} \quad c = -4a$$

which are both even

(b)(i)
E2

$$\rho = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$$

$$\rho^7 = \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \right)^7$$

$$= \cos 2\pi + i \sin 2\pi$$

$$= 1$$

$\therefore \rho$ is a root of $x^7 - 1 = 0$ i.e. ρ is a root of

$$(x - 1)(x^6 + x^5 + x^4 + x^3 + x^2 + x + 1) = 0$$

but $x \neq 0$, so ρ satisfies

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$$

$$\text{i.e. } 1 + \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 = 0$$

OR...

$1 + \rho + \rho^2 + \dots + \rho^6$ is a GP with $a = 1$

$$r = \rho$$

$$n = 7$$

$$\text{so, } 1 + \rho + \rho^2 + \dots + \rho^6 = \frac{1(\rho^7 - 1)}{\rho - 1}$$

$$= \frac{(\text{cis } \frac{2\pi}{7})^7}{\rho - 1}$$

$$= \frac{\cos 2\pi + i \sin 2\pi - 1}{\rho - 1} = 0$$

2 marks : correct solution

OR

1 mark : substantially correct

2 marks : correct solution

OR

1 mark : substantially correct

*Note: this must be proven/shown.
Marks not awarded for using
calculator*

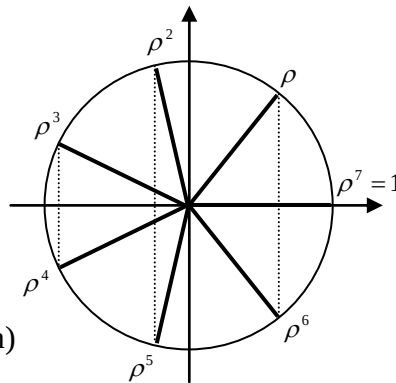
Q.5 pg3

(b)(ii)
E2

Since the coefficients of $x^2 + ax + b = 0$ are real, and $\alpha = \rho + \rho^2 + \rho^4$ is a complex root, then $\bar{\alpha}$ must also be a root.

$$\begin{aligned}\therefore \beta = \bar{\alpha} &= \overline{\rho + \rho^2 + \rho^4} \\ &= \bar{\rho} + \bar{\rho^2} + \bar{\rho^4} \\ &= \rho^6 + \rho^5 + \rho^3\end{aligned}$$

Since $\bar{\rho} = \rho^6$, $\bar{\rho^2} = \rho^5$
and $\bar{\rho^4} = \rho^3$ (see diagram)



2 marks : correct solution

OR

1 mark : substantially correct

(b)(iii)
E2, E4

sum of roots:

$$\alpha + \beta = -a$$

$$\text{ie } a = -(\alpha + \beta)$$

$$\begin{aligned}&= -(\rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6) \text{ from (ii)} \\ &= 1 \text{ from (i)}\end{aligned}$$

product of roots:

$$\begin{aligned}b &= \alpha\beta \\ &= (\rho + \rho^2 + \rho^4)(\rho^3 + \rho^5 + \rho^6) \\ &= \rho(1 + \rho + \rho^3)\rho^3(1 + \rho^2 + \rho^3) \\ &= \rho^4 \left(\underbrace{1 + \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6}_{\downarrow 0} + 2\rho^3 \right) \\ &= \rho^4 (0 + 2\rho^3) \\ &= 2\rho^7 \\ &= 2 \quad (\rho^7 = 1)\end{aligned}$$

$$\text{i.e. } a = 1 \quad \& \quad b = 2$$

2 marks : correct solution

(both **a** and **b** correct)

OR

1 mark : partially correct

(either **a** or **b** correct)

(b)(iv)
E2

from (iii), the quadratic equation is:

$$x^2 + x + 2 = 0$$

$$\begin{aligned}\text{so... } x &= \frac{-1 \pm \sqrt{1-8}}{2} \\ &= \frac{-1 \pm i\sqrt{7}}{2}\end{aligned}$$

$$\therefore \text{Im}(\alpha) = \frac{\pm\sqrt{7}}{2}, \quad \text{but } \alpha = \rho + \rho^2 + \rho^4$$

and $\text{Im}(\alpha) > 0$, from diagram in (ii)

$$\therefore \text{Im}(\alpha) = \frac{\sqrt{7}}{2}$$

Q.5 pg3

$$\begin{aligned}\text{Now, } \operatorname{Im}(\alpha) &= \operatorname{Im}(\rho + \rho^2 + \rho^4) \\ &= \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} \\ &= \sin \frac{2\pi}{7} + \sin \left(\pi - \frac{3\pi}{7} \right) + \sin \left(\pi + \frac{\pi}{7} \right) \\ &= \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} \\ &= \frac{\sqrt{7}}{2}, \text{ as required}\end{aligned}$$

2 marks : correct solution

OR

1 mark : substantially correct

Note: this must be deduced, ie from earlier parts. Marks not awarded for using calculator